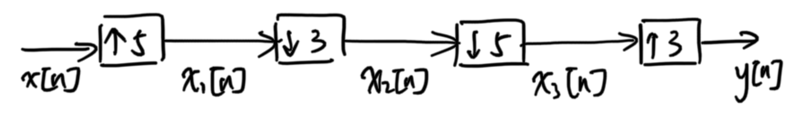
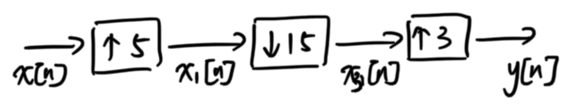
98

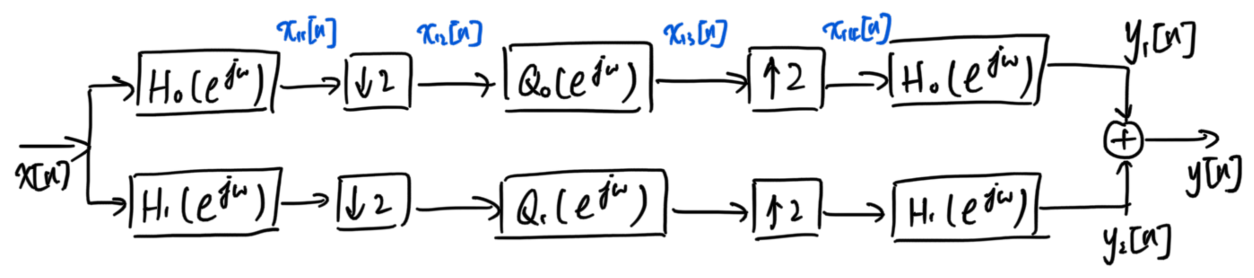
4.33





$$y[n]=\begin{cases} x_3[n/3], & 3\mid n \\ 0, & 3\nmid n\end{cases} = \begin{cases} x_1[5n/3], & 3\mid n \\ 0, & 3\nmid n\end{cases} = \begin{cases} x_1[5n], & 3\mid n \\ 0, & 3\nmid n\end{cases} = \begin{cases} x[n], & 3\mid n \\ 0, & 3\nmid n\end{cases}$$

4.49



$$Y_1(e^{j\omega}) = H_0(e^{j\omega})X_{14}(e^{j\omega}) = H_0(e^{j\omega})X_{13}(e^{j2\omega})$$

$$= H_0(e^{j\omega})Q_0(e^{j2\omega})X_{12}(e^{j2\omega})$$

$$= H_0(e^{j\omega})Q_0(e^{j2\omega})\cdot\frac{1}{2}\sum_{i=0}^{1}X_{11}(e^{j(\frac{\omega-2\pi i}{2})})$$

$$= \frac{1}{2}H_0(e^{j\omega})Q_0(e^{j2\omega})\sum_{i=0}^{1}X(e^{j(\frac{\omega-2\pi i}{2})})H_0(e^{j(\frac{\omega-2\pi i}{2})})$$

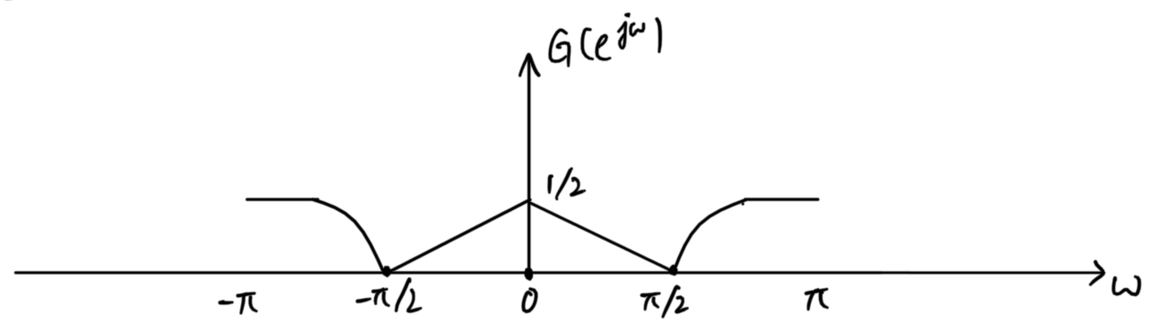
同理可得

$$Y_2(e^{j\omega}) = \frac{1}{2}H_1(e^{j\omega})Q_1(e^{j2\omega})\sum_{i=0}^{1}X(e^{j(\frac{\omega-2\pi i}{2})})H_1(e^{j(\frac{\omega-2\pi i}{2})})$$

令 $x[n]=\delta[n] \Leftrightarrow X(e^{j\omega})=1$

则 $\sum_{i=0}^{1}H_0(e^{j(\frac{\omega-2\pi i}{2})}) = \sum_{i=0}^{1}H_1(e^{j(\frac{\omega-2\pi i}{2})}) = 1$

因此 $G(e^{j\omega}) = \frac{1}{2}\left[H_0(e^{j\omega})Q_0(e^{j2\omega}) + H_1(e^{j\omega})Q_1(e^{j2\omega})\right]$

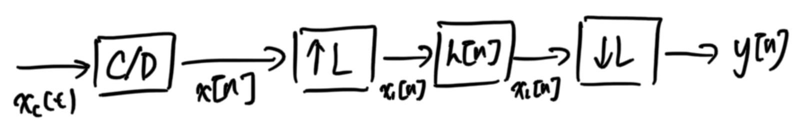


4.52 $H(e^{j\omega}) = \begin{cases} e^{-j\omega}, & |\omega|<\pi/L \\ 0 & \pi/L<|\omega|\le\pi\end{cases} \Leftrightarrow h[n] = \frac{1}{2\pi}\int_{-\pi/L}^{\pi/L}e^{-j\omega}\cdot e^{j\omega n}d\omega$

$$= \frac{1}{2\pi}\int_{-\pi/L}^{\pi/L}e^{j\omega(n-1)}d\omega$$

$$= \frac{1}{2\pi}\cdot\frac{1}{j(n-1)}\cdot 2j\sin\left(\frac{\pi(n-1)}{L}\right)$$

$$= \frac{\sin(\pi(n-1)/L)}{\pi(n-1)} = \frac{1}{L}Sa\left(\frac{n-1}{L}\right)$$



$$y[n] = x_2[Ln] = \sum_{k=-\infty}^{+\infty}x_1[k]\cdot\frac{1}{L}Sa\left(\frac{Ln-k-1}{L}\right)$$

$$= \sum_{m=-\infty}^{+\infty}x_1[mL]\cdot\frac{1}{L}Sa\left(\frac{Ln-mL-1}{L}\right)$$

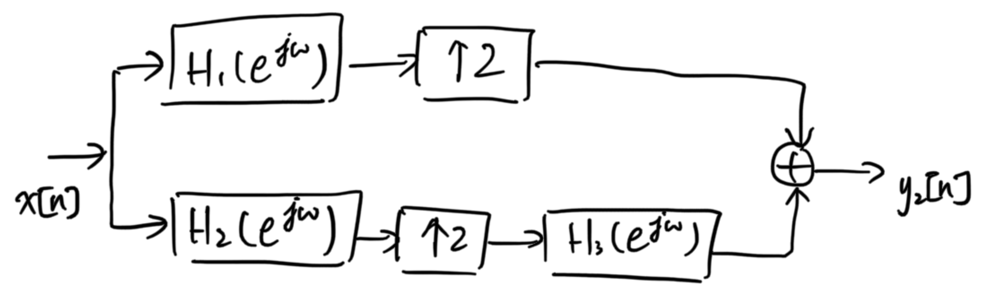
$$= \sum_{m=-\infty}^{+\infty}x_1[m]\cdot\frac{1}{L}Sa\left(n-m-\frac{1}{L}\right)$$

$$= \frac{1}{L}\sum_{m=-\infty}^{+\infty}x_c(mT)\cdot\frac{\sin(\pi(n-m-\frac{1}{L}))}{\pi(n-m-\frac{1}{L})}$$

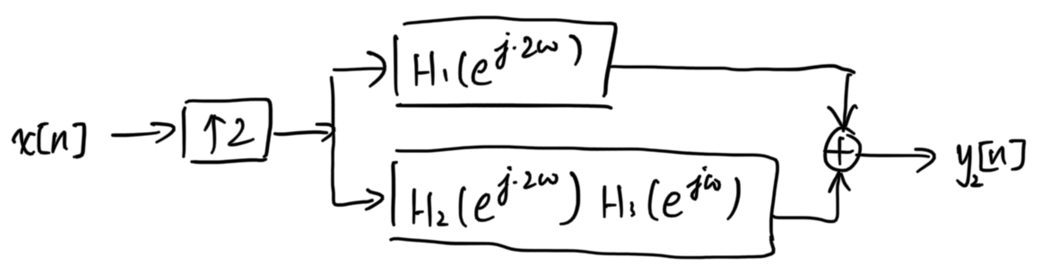
$$= \frac{(-1)^{n+1}}{L}\cdot\sin\left(\frac{\pi}{L}\right)\sum_{m=-\infty}^{+\infty}x_c(mT)\cdot\frac{(-1)^m}{\pi(n-m-\frac{1}{L})}$$

$$= \frac{1}{L}x_c[nT-\frac{T}{L}]$$

4.58 (a)



$$\Updownarrow$$



记 $h(i,j) = h_i[j]$，其中 $i=1,2,3$，$j=0,1,2$

$$H_1(e^{j2\omega}) \Leftrightarrow h_1^{(2)}[n] = \begin{cases} h(1,n/2), & n=0,2,4 \\ 0, & \text{otherwise}\end{cases}$$

$$H_2(e^{j2\omega}) \Leftrightarrow h_2^{(2)}[n] = \begin{cases} h(2,n/2), & n=0,2,4 \\ 0, & \text{otherwise}\end{cases}$$

$$H_2(e^{j2\omega})H_3(e^{j\omega}) \Leftrightarrow h_{23}[n] = \sum_{k=-\infty}^{+\infty}h_2^{(2)}[k]h_3[n-k]$$

$$= \sum_{k=0}^{2}h_2^{(2)}[2k]h_3[n-2k]$$

$$= \sum_{k=0}^{2}h_2[k]h_3[n-2k]$$

令 $h_1[0:2]=[a,c,e] \Leftrightarrow h_1^{(2)}[0:4]=[a,0,c,0,e]$

$h_2[0:2]=[b,d,0] \Leftrightarrow h_2^{(2)}[0:2]=[b,0,d]$

$h_3[0:2]=[0,1,0]=\delta[n-1]$

$$\begin{cases} h_{23}[0] = h_2[0]h_3[0] \ ✓ \\ h_{23}[1] = h_2[0]h_3[1] \\ h_{23}[2] = h_2[0]h_3[2]+h_2[1]h_3[0] \ ✓ \\ h_{23}[3] = h_2[1]h_3[1] \\ h_{23}[4] = h_2[1]h_3[2]+h_2[2]h_3[0] \ ✓ \\ h_{23}[5] = h_2[2]h_3[1] \ ✓ \\ h_{23}[6] = h_2[2]h_3[2] \ ✓ \\ h_{23}[n] = 0, \quad n\ge 7 \text{ 或 } n\le -1\end{cases}$$

$$\begin{cases} h_1^{(2)}[0] = h_1[0] \\ h_1^{(2)}[2] = h_1[1] \\ h_1^{(2)}[4] = h_1[2] \\ h_1^{(2)}[n] = 0, \text{ otherwise}\end{cases}$$

(b) 图P4.58-1中，乘法次数 $= 5\times2 = 10$ 次

图P4.58-2中，乘法次数 $= 3\times2+3 = 9$ 次

$h_3[n]$中乘1不算乘法

-2

4.63 (a) $\mathcal{E}\{e[n]\} = \int_{-\Delta/2}^{\Delta/2}e\cdot\frac{1}{\Delta}de = 0$

$\mathrm{var}\{e[n]\} = \mathcal{E}\{e[n]^2\} - \mathcal{E}\{e[n]\}^2$

$= \int_{-\Delta/2}^{\Delta/2}e^2\cdot\frac{1}{\Delta}de = \frac{1}{3\Delta}\cdot2\times\left(\frac{\Delta}{2}\right)^3 = \frac{\Delta^2}{12} = \sigma_e^2$

$$\begin{cases} R_{ee}[n] = \mathcal{E}\{e[m]e[m+n]\}, \quad n\ne0 \\ \quad = \int_{-\Delta/2}^{\Delta/2}\int_{-\Delta/2}^{\Delta/2}ee'\cdot\left(\frac{1}{\Delta}\right)^2de\,de' \\ \quad = \mathcal{E}\{e[m]\}\mathcal{E}\{e[m+n]\} = 0 \\ R_{ee}[0] = \mathcal{E}\{e[m]^2\} \\ \quad = \frac{\Delta^2}{12}\end{cases} \Rightarrow R_{ee}[n] = \frac{\Delta^2}{12}\delta[n]$$

(b) $SNR = \frac{\sigma_x^2}{\sigma_e^2} = \frac{12\sigma_x^2}{\Delta^2}$

(c) $\mathcal{E}\{x[n]*h[n]\} = \mathcal{E}\{x[n]\}*h[n] = 0$

$\mathcal{E}\{e[n]*h[n]\} = \mathcal{E}\{e[n]\}*h[n] = 0$

$\mathcal{E}\{(x[n]*h[n])^2\} = \mathcal{E}\{(\sum_{k=0}^{\infty}\frac{1}{2}(a^k+(-a)^k)x[n-k])^2\}$

$= \mathcal{E}\{(\sum_{k=0}^{\infty}a^{2k}x[n-2k])^2\}$

$= \mathcal{E}\{\sum_{k=0}^{\infty}\sum_{r=0}^{\infty}a^{2(k+r)}x[n-2k]x[n-2r]\}$

$= \sum_{k=0}^{\infty}\sum_{r=0}^{\infty}a^{2(k+r)}\sigma_x^2\delta[2k-2r]$

$= \sum_{k=0}^{\infty}a^{4k}\sigma_x^2$

$= \frac{\sigma_x^2}{1-a^4}$

$\mathcal{E}\{(e[n]*h[n])^2\} = \frac{\sigma_e^2}{1-a^4} = \frac{\Delta^2}{12(1-a^4)}$

$SNR = \frac{\sigma_x^2/(1-a^4)}{\sigma_e^2/(1-a^4)} = \frac{12\sigma_x^2}{\Delta^2}$

(d) $f[n] = x[n]e[n]$

$\mathcal{E}\{f[n]\} = \mathcal{E}\{x[n]e[n]\} = 0$

$\mathrm{var}\{f[n]\} = \mathcal{E}\{x^2[n]e^2[n]\} = \mathcal{E}\{x[n]^2\}\mathcal{E}\{e[n]^2\}$

$= \sigma_x^2\sigma_e^2 = \frac{\Delta^2}{12}\sigma_x^2 = \sigma_f^2$

$R_{ff}[n] = \mathcal{E}\{f[m]f[m+n]\}$

$= \mathcal{E}\{x[m]x[m+n]e[m]e[m+n]\}$

$= R_{xx}[n]R_{ee}[n]$

$= \sigma_x^2\sigma_e^2\delta^2[n]$

$= \frac{\Delta^2}{12}\sigma_x^2\delta[n]$

(e) $SNR = \frac{\sigma_x^2}{\sigma_f^2} = \frac{1}{\sigma_e^2} = \frac{12}{\Delta^2}$

(f) $\mathcal{E}\{f[n]*h[n]\} = 0$

$\mathcal{E}\{(f[n]*h[n])^2\} = \frac{\sigma_f^2}{1-a^4} = \frac{\Delta^2}{12}\cdot\frac{\sigma_x^2}{1-a^4}$

$\mathcal{E}\{(x[n]*h[n])^2\} = \frac{\sigma_x^2}{1-a^4}$

$SNR = \frac{\sigma_x^2/(1-a^4)}{\sigma_f^2/(1-a^4)} = \frac{12}{\Delta^2}$